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Dooyoung Kim

Current appointment

From Jun 2026 **Lecturer in Computer Science and Information Technology**,
Continuing academic appointment in research and teaching.
School of Computing, Engineering and Mathematical Sciences,
La Trobe University, Melbourne, Australia.

Research interests

Areas: Human–computer interaction; spatial computing; augmented and virtual reality; human–AI interaction; remote collaboration and telepresence.

Research focus: Spatial interfaces and representations for sharing human experiences and expertise across space and time.

Education

Mar 2021 – Feb 2024 **Ph.D. in Culture Technology**, KAIST, Daejeon, Korea.
Thesis: *Space-Adaptive Mutual Space Generation for Mixed Reality Remote Collaboration.*
UVR Lab. · Advisor: Prof. Woontack Woo.

Mar 2019 – Feb 2021 **M.S. in Culture Technology**, KAIST, Daejeon, Korea.
Thesis: *Adjusting Relative Translation Gains According to Space Size in Redirected Walking for Mixed Reality Mutual Space Generation.*
UVR Lab. · Advisor: Prof. Woontack Woo.
Visiting Researcher (Dec 2019 – Feb 2020): Imagine Lab, École des Ponts ParisTech, France (Prof. Vincent Lepetit).

Mar 2014 – Feb 2019 **B.S. in Mechanical Engineering**, KAIST, Daejeon, Korea.
Exchange Student (Sep-Dec 2016): Tsinghua University, Beijing, China.
Lab Interns: Hubo Lab - Humanoid Robotics (Jan-Mar 2017);
UVR Lab - AR/VR (Jun-Aug 2018).

Previous appointments

Mar 2024 – May 2026 **Senior Researcher**, KAIST, Daejeon, Korea.
Co-PI on Meta-Object (US$3M total project funding); contributed to proposal preparation and grant acquisition, with decision authority over research-equipment purchases. System Integration Manager on OpenXR (US$6M total project funding); integrated research outputs across multiple KAIST teams. Co-advised and mentored 7 graduate students (PhD/MS), with publications at IEEE TVCG, ISMAR, VR.

Jan 2025 – Dec 2025 **NYU–KAIST Postdoc Collaboration**, New York University, NY, USA.
NYU Future Reality Lab (Prof. Ken Perlin); on-site and remote visits throughout 2025 on the KAIST–NYU Meta-Museum initiative.

May 2022 – Mar 2023 **AR/VR Researcher**, Korea Institute of Science and Technology Information (KISTI), Daejeon, Korea.
Joint research with KAIST on the OpenXR: Untact Realistic Platform Technology government project.

Apr 2021 – Nov 2022 **Creator**, ZER01NE program, Hyundai Motor Group (HMG).
Selected as AR/VR researcher. Projects: *Automatic SONATA* — scenery style-transfer system for autonomous vehicles (2021); *Holo-bot* — XR social interface robot with holographic avatar (2022).

Technical expertise

Platforms: Meta Quest 3/Pro, HoloLens 2, Apple Vision Pro, custom CAVE/wall-display setups, OpenXR.

Software	Unity (C#), Unreal, ARKit/ARCore, MRTK; Python (PyTorch, NumPy); OpenCV, COLMAP/SLAM; human-subject statistics.
Methods	XR prototyping, spatial AI / scene understanding, semantic scene graphs, 3D reconstruction, world-model priors, redirected walking, perceptual experiments, IRB-approved human-subject studies, mixed-methods evaluation.

Awards and honours

Oct 2025	**Best Paper Award** (Top 1%), IEEE ISMAR 2025 — *Viewpoint-Tolerant Depth Perception for Shared Extended Space Experience on Wall-Sized Display.*
Oct 2025	**Best Paper Award** (Top 1%), IEEE ISMAR 2025 — *Visuo-Tactile Feedback with Hand Outline Styles for Modulating Affective Roughness Perception.*
Jan 2025	NYU–KAIST Postdoc Collaboration Grant, New York University.
Oct 2024	**Best Paper Award** (Top 1%), IEEE ISMAR 2024 — *Spatial Affordance-Aware Interactable Subspace Allocation for Mixed Reality Telepresence.*
Mar 2023	KAIST Start-up Scholarship, KAIST Alumni Academic Scholarship Foundation.
Dec 2022	**Best Presentation Award** (1st Prize), The Asia-Pacific Workshop on Mixed and Augmented Reality (APMAR) 2022 — Springer *Virtual Reality* paper invited and presented.
Apr 2022	NCSOFT Ph.D. Scholarship.
May 2021	**Honorable Mention Award** (Top 5%), ACM CHI 2021.

Grants and research projects

Jan 2025 – Dec 2025	**Meta-Museum** — NYU–KAIST Collaborative Research Grant. *Tele-experience system for a meta-museum between Daejeon (Korea) and New York (USA).* Role: Project Leader. Collaboration with NYU Future Reality Lab (Prof. Ken Perlin).
Apr 2024 – May 2026	**Meta-Object** — Korean Government (IITP) national research project. *Real-time XR interface technology for environmental adaptation; object modelling, multimodal feedback, intelligent platform.* Project period: Mar 2024 – Dec 2026. Role: Co-Principal Investigator (Co-PI; PI: Prof. Woontack Woo). Contributed to proposal preparation and grant acquisition; exercised decision authority over research-equipment purchases. Directed 40 researchers across 8 multidisciplinary teams. Total project funding: US$3M.
Dec 2021 – May 2026	**OpenXR: TranSpace 3.0** — Korean Government (NST) national research project. *Development of untact realistic OpenXR remote collaboration platform technology.* Role: System Integration Manager; integration lead for research outputs across multiple KAIST teams. Total project funding: US$6M over 6 years.
Sep 2019 – Dec 2020	**TranSpace 2.0** — Korean Government (NRF) national research project. *User restoration technology for ultra-realistic remote virtual interaction.* Role: Research Assistant. AR/VR remote-collaboration system and VR locomotion research.
Apr 2020 – Dec 2020	**National R&D Real Challenge** — Korean Government (KIRD). *Reinforcement of visual cognitive abilities for AR/VR/MR remote collaboration.* Role: Research Assistant. Visual-perception research for VR remote-collaboration scenarios.

Publications

Summary	31 refereed publications in total — 12 journal ([J.1–J.12]), 9 conference ([C.1–C.9]), and 10 workshop / adjunct / demo ([W.1–W.10]). All listed below are peer-reviewed.

Bold indicates Dooyoung Kim; ★ *marks award-winning papers. Co-advised students' first-author papers are labelled* (co-advised, student first author).

Refereed journal papers

[J.12] Kim, S.-Y., **Kim, D.**, Son, T., Kim, Y. B., Cho, W., & Woo, W. (in press). Mix3R: Zero-shot interactive 3D scene reconstruction from a single image in mixed reality. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*. *(co-advised, student first author)*

[J.11] Kim, S., **Kim, D.**, Choi, S., & Woo, W. (2026). Spatial affordance-aware affine transformation between heterogeneous spaces for mixed reality remote collaboration. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, *32*(5), 3905–3913. https://doi.org/10.1109/TVCG.2026.3680606 *(co-advised, student first author)*

[J.10] Kim, S.-Y., **Kim, D.**, Cho, W., Song, H., Kang, S., & Woo, W. (2026). SceneLinker: Compositional 3D scene generation via semantic scene graph from RGB sequences. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, *32*(5), 3125–3135. https://doi.org/10.1109/TVCG.2026.3679893 *(co-advised, student first author)*

[J.9] Choi, S., **Kim, D.**, Ha, T., Kim, S., & Woo, W. (2026). Task breakpoint generation using origin-centric graph in virtual reality recordings for adaptive playback. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, *32*(5), 3283–3292. https://doi.org/10.1109/TVCG.2026.3679910 *(co-advised, student first author)*

[J.8] Ha, T., Cho, W., **Kim, D.**, & Woo, W. (2026). Int3DNet: Scene-motion cross attention network for 3D intention prediction in mixed reality. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, *32*(5), 4185–4195. https://doi.org/10.1109/TVCG.2026.3679899 *(co-advised, student first author)*

[J.7] **Kim, D.**, Hong, J., Ko, H., & Woo, W. (2025). Viewpoint-tolerant depth perception for shared extended space experience on wall-sized display. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, *31*(11), 9932–9942. https://doi.org/10.1109/TVCG.2025.3616758 ★ Best Paper Award (Top 1%), IEEE ISMAR 2025

[J.6] Baeck, M., Shin, Y., **Kim, D.**, Lee, H., Yoon, S. H., & Woo, W. (2025). Visuotactile feedback with hand outline styles for modulating affective roughness perception. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, *31*(11), 10099–10108. https://doi.org/10.1109/TVCG.2025.3616805 *(co-advised, student first author)* ★ Best Paper Award (Top 1%), IEEE ISMAR 2025

[J.5] **Kim, D.**, Ha, T., Hong, J., Kim, S., Choi, S., Ko, H., & Woo, W. (2025). Meta-Objects: Interactive and multisensory virtual objects learned from the real world for use in augmented reality. *IEEE Computer Graphics and Applications (CG&A)*, *45*(3), 134–143. https://doi.org/10.1109/MCG.2025.3555901

[J.4] **Kim, D.**, Kim, S., Shin, J.-E., Yoon, B., Kim, J., Lee, J., & Woo, W. (2023). The effects of spatial configuration on relative translation gain thresholds in redirected walking. *Virtual Reality*, *27*(2), 1233–1250. https://doi.org/10.1007/s10055-022-00734-3 ★ Best Presentation Award (1st Prize), APMAR 2022

[J.3] Kim, J., Jang, H., **Kim, D.**, & Lee, J. (2023). Exploration of the virtual reality teleportation methods using hand-tracking, eye-tracking, and EEG. *International Journal of Human-Computer Interaction (IJHCI)*, *39*(20), 4112–4125. https://doi.org/10.1080/10447318.2022.2109248

[J.2] Yoon, B., Shin, J.-E., Kim, H.-I., Oh, S. Y., **Kim, D.**, & Woo, W. (2023). Effects of avatar transparency on social presence in task-centric mixed reality remote collaboration. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, *29*(11), 4578–4588. https://doi.org/10.1109/TVCG.2023.3320258

[J.1] Shin, J.-E., Yoon, B., **Kim, D.**, & Woo, W. (2023). The effects of spatial complexity on narrative experience in space-adaptive AR storytelling. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, *29*(12), 5137–5148. https://doi.org/10.1109/TVCG.2022.3201934

Refereed conference papers

[C.9] Kim, S.-Y., Elskhawy, A., Ha, T., **Kim, D.**, Shin, E., Busam, B., & Woo, W. (2026). DeWorldSG: Depth-aware 3D semantic scene graph generation via world-model priors. In *Computer Vision – ECCV 2026* (Lecture Notes in Computer Science, Vol. 17051, pp. 1–19). Springer. https://doi.org/10.1007/978-3-032-37356-4_1 *(co-advised, student first author)*

[C.8] Choi, S., Bae, S. A., **Kim, D.**, & Woo, W. (in press). Object-driven event retrieval with assembled object interface for mixed-reality assembly playback. In *2026 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE. *(co-advised, student first author)*

[C.7] **Kim, D.**, Kim, S., Choi, S., & Woo, W. (2024). Spatial affordance-aware interactable subspace allocation for mixed reality telepresence. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* (pp. 1256–1265). IEEE. https://doi.org/10.1109/ISMAR62088.2024.00142 ★ Best Paper Award (Top 1%)

[C.6] Kim, S., **Kim, D.**, Shin, J.-E., & Woo, W. (2024). Object cluster registration of dissimilar rooms using geometric spatial affordance graph to generate shared virtual spaces. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)* (pp. 796–805). IEEE. https://doi.org/10.1109/VR58804.2024.00099 *(co-advised, student first author)*

[C.5] **Kim, D.**, & Woo, W. (2023). Edge-centric space rescaling with redirected walking for dissimilar physical-virtual space registration. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* (pp. 829–838). IEEE. https://doi.org/10.1109/ISMAR59233.2023.00098

[C.4] **Kim, D.**, Kim, J., Shin, J.-E., Yoon, B., Lee, J., & Woo, W. (2022). Effects of virtual room size and objects on relative translation gain thresholds in redirected walking. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)* (pp. 379–388). IEEE. https://doi.org/10.1109/VR51125.2022.00057

[C.3] Shin, J.-E., Yoon, B., **Kim, D.**, Kim, H.-I., & Woo, W. (2022). The effects of device and spatial layout on social presence during a dynamic remote collaboration task in mixed reality. In *2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* (pp. 394–403). IEEE. https://doi.org/10.1109/ISMAR55827.2022.00055

[C.2] **Kim, D.**, Shin, J.-E., Lee, J., & Woo, W. (2021). Adjusting relative translation gains according to space size in redirected walking for mixed reality mutual space generation. In *2021 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)* (pp. 653–660). IEEE. https://doi.org/10.1109/VR50410.2021.00091

[C.1] Shin, J.-E., Yoon, B., **Kim, D.**, & Woo, W. (2021). A user-oriented approach to space-adaptive augmentation: The effects of spatial affordance on narrative experience in an augmented reality detective game. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI)* (Article 722, pp. 1–13). Association for Computing Machinery. https://doi.org/10.1145/3411764.3445675 ★ Honorable Mention Award (Top 5%)

Refereed workshop, adjunct, and demo papers

[W.10] **Kim, D.**, & Woo, W. (2025). Spatio-temporal mixed and augmented reality experience description for interactive playback. In *2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 488–491). IEEE. https://doi.org/10.1109/ISMAR-Adjunct68609.2025.00098

[W.9] **Kim, D.**, Kim, K., Silva, C. T., Sun, Q., Turakhia, D., Wang, Z., Wang, K., Perlin, K., Feiner, S., & Woo, W. (2025). An overview of the 1st International Workshop on Spatial Memory in XR: The future of memory capture and replay through XR and AI (XRMemory). In *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 1132–1133). IEEE. https://doi.org/10.1109/VRW66409.2025.00224

[W.8] Kim, S., **Kim, D.**, Son, T., & Woo, W. (2025). RealityCrafter: User-guided editable 3D scene generation from a single image in mixed reality. In *Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct)* (Article 109, pp. 1–3). Association for Computing Machinery. https://doi.org/10.1145/3746058.3758405 *(co-advised, student first author)*

[W.7] Ha, T., Choi, S., Kim, S., **Kim, D.**, & Woo, W. (2025). Human-scene interaction data generation with virtual environment using user-centric scene graph. In *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 1422–1423). IEEE. https://doi.org/10.1109/VRW66409.2025.00357 *(co-advised, student first author)*

[W.6] Kim, J., **Kim, D.**, Kim, B., Kim, H., & Lee, J. (2023). Holobot: Hologram based extended reality telepresence robot. In *Companion Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI Companion)* (pp. 60–64). Association for Computing Machinery. https://doi.org/10.1145/3568294.3580043

[W.5] **Kim, D.**, Kim, H.-I., & Woo, W. (2022). Mutual space generation with relative translation gains in redirected walking for asymmetric remote collaboration. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 674–677). IEEE. https://doi.org/10.1109/ISMAR-Adjunct57072.2022.00140

[W.4] Du, J., Park, S., Yum, J., Özdemir, Z. Ö., **Kim, D.**, Oh, S. Y., & Yoon, S. H. (2022). Art Rich: Place your AR artwork. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 667–670). IEEE. https://doi.org/10.1109/ISMAR-Adjunct57072.2022.00138

[W.3] Kim, S., Jang, H., Oh, K. T., Oh, S. Y., **Kim, D.**, Woo, W., Lee, J., Ahn, J., & Yoon, S. H. (2022). Bring store in my room: AR store authoring system for spatial experience in mobile shopping. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 654–656). IEEE. https://doi.org/10.1109/ISMAR-Adjunct57072.2022.00135

[W.2] Lee, S. U., Yun, J., Kim, D., **Kim, D.**, Oh, S. Y., & Yoon, S. H. (2022). CARDS: Comprehensive AR docent system. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 739–743). IEEE. https://doi.org/10.1109/ISMAR-Adjunct57072.2022.00156

[W.1] Kim, H.-I., Kim, T., Song, E., Oh, S. Y., **Kim, D.**, & Woo, W. (2021). Multi-scale mixed reality collaboration for digital twin. In *2021 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 435–436). IEEE. https://doi.org/10.1109/ISMAR-Adjunct54149.2021.00098

Patents and standardisation

Granted patents

2026 [US] Edge-centric space rescaling method for dissimilar space registration and system therefor (US12555342B2).

2025 [US/KR] Virtual reality space adjusting method with relative translation gain in redirected walking and the system thereof (US12333628B2; KR102829349B1).

2024 [KR] Method and system for generating remote collaboration mutual space (KR102728698B1).

Filed patent applications

Years indicate filing dates.

2026 [KR] Method and system for processing mixed and augmented reality experiences. *Filed; not yet published.*

2026 [KR] Image processing device and recording medium for storing video processed therefrom. *Filed; not yet published.*

2026 [KR] A system for real-time affordance visualization and interaction with meta-objects based on dynamic user context. *Filed; not yet published.*

2024 [KR] Method of operating mixed reality telepresence system and mixed reality telepresence system performing the same (app. 10-2024-0005235; pub. KR20250110967A).

2024 [KR] Spatial extension using human stereoscopic perception with a large wall display (app. 10-2024-0184973).

2024 [KR] Object cluster registration of dissimilar rooms using geometric spatial affordance graph to generate shared virtual spaces (app. 10-2024-0165710).

2023 [KR] Edge-centric space rescaling method for dissimilar space registration and the system thereof (app. 10-2023-0100226; pub. KR20250019288A).

Published PCT applications

Years indicate filing dates.

2025 [PCT] Method for operating mixed reality telepresence system and mixed reality telepresence system performing same (PCT/KR2025/000529; pub. WO2025150937A1).

2023 [PCT] Method and system for generating shared space for remote collaboration (PCT/KR2023/016052; pub. WO2024219569A1).

International standardisation

2025–present Project Editor, ISO/IEC AWI 26073 (ISO/IEC JTC 1/SC 24).
Spatio-Temporal Mixed and Augmented Reality Experience Description (MAR-ED) for Interactive Playback. (Under development.)

Invited talks

2026 University of Melbourne (host: Prof. Taehyun Rhee and Assoc. Prof. Wafa Johal; School of Computing and Information Systems) — Connecting Intelligences beyond Boundaries.

2026 University of Bologna (host: Asst. Prof. Pasquale Cascarano; Department of the Arts) — Connecting Intelligences beyond Boundaries.

2026 Monash University (host: Prof. Tim Dwyer; Department of Human Centred Computing) — Connecting Intelligences beyond Boundaries.

2026 University of Adelaide (host: Assoc. Prof. Gun Lee; Australian Research Centre for Interactive and Virtual Environments) — Interactive XR Intelligence.

2026 Handong Global University (host: Prof. Minkyu Ahn; School of Artificial Intelligence, Computer and Electrical Engineering) — XRMemory and Physical AI.

2024 Korea Computer Congress (KCC), Top-Conference Session.

2024 University of Minnesota (host: Prof. Victoria Interrante) — Redirected Walking & Spatial AI.

2023 Korea Computer Congress (KCC), Top-Conference Session.

2023 Korea Software Congress (KSC), Top-Conference Session.

2023 HCI Korea Conference — Metaverse Research and Development Session.

2022 ACM SIGGRAPH Asia — OpenXR BoF Session.

2022 Asia-Pacific Workshop on Mixed and Augmented Reality (APMAR), Pitch-Your-Work Session.

Conference and editorial service

Editorial

2025– Lead Guest Editor, Special Issue on "XRMemory: The Future of Memory Capture, Reconstruction, and Replay in XR," Springer *Virtual Reality* Journal (2025 JIF: 5.4).

2025– Guest Editor, Special Issue on "XR for Knowledge Work and Workplace Enhancement: Human-Centered, Productive, and Ethical XR Workspaces," Springer *Journal of Ambient Intelligence and Humanized Computing*.

Organising committee — IEEE ISMAR / VR (flagship XR venues)

2027 Doctoral Consortium Chair, IEEE VR 2027.

2027 Principal Organiser, 5th International XRMemory Workshop @ IEEE VR 2027.

2026 Industrial Exhibition & Sponsorship Chair, IEEE ISMAR 2026.

2026 Best Paper Award Committee, IEEE ISMAR 2026.

2026 Best Demo Award Committee, IEEE ISMAR 2026.

2026 Principal Organiser, 4th International XRMemory Workshop @ IEEE ISMAR 2026.

2026 XR Standardization Committee, IEEE VR 2026.

2026 Principal Organiser, 3rd International XRMemory Workshop @ IEEE VR 2026.

2025 Assistant to General Chairs & Social Event Chair, IEEE ISMAR 2025 Organising Committee.

2025 Principal Organiser, 2nd International XRMemory Workshop @ IEEE ISMAR 2025.

2025 Organiser, XRWorks'25 Workshop @ IEEE ISMAR 2025; IPC, XRStand Workshop.

2025 Principal Organiser, 1st International XRMemory Workshop @ IEEE VR 2025.

Organising committee — other

2025 Organiser & Program Chair, 3rd International Symposium on Meta-AI for X (ISMAX).

2025 Organiser, MetaMuseum Workshop @ HCI Korea Conference.

2024 Poster Chair (Computer Graphics & Interaction); Metaverse+AI+XR Workshop Organiser — Korea Computer Congress (KCC).

2024 Poster Chair (CG/HCI/AI/Metaverse/AR/VR) — Korea Software Congress (KSC).

2024 Organiser & Web Chair, 2nd International Symposium on Meta-AI for X (ISMAX).

2023 Organiser & Student Chair, 1st International Symposium on Meta-AI for X (ISMAX).

International Program Committee (IPC) & Reviewer

IEEE VR 2027 (IPC), 2026 (IPC), 2025 (IPC), 2024, 2023, 2022.

IEEE ISMAR 2026 (IPC), 2025 (IPC), 2024, 2023, 2022, 2021.

ACM VRST 2026 (IPC), 2025 (IPC), 2024 (IPC).

ACM CHI 2027, 2026, 2025, 2024, 2023.

IEEE/ACM HRI 2023.

Teaching

Lecturer — La Trobe

2026 CSE2002 — Agentic AI Software Engineering; coordinator; planned T1 2027.

Teaching assistant — KAIST

Mar-Jun 2025 GCT700 — AR project mentoring; teaching assistant on Spatial AI & XRMemory.

Sep-Dec 2024 MV600 — AR/Metaverse project mentoring; teaching assistant on MR telepresence.

Mar-Jun 2024 GCT700 — AR project mentoring; teaching assistant on XR & Spatial Computing.

Mar-Jun 2023	CTP445 — AR fundamentals; discussion sessions; teaching assistant on SLAM with OpenCV.
Sep-Dec 2022	GCT555 — Unity & AR/VR for graduate students; teaching assistant on HoloLens 2, MRTK with Unity.
Mar-Jun 2022	CTP445 — Unity & AR/VR for graduate students; teaching assistant on smartphone AR with ARCore and Unity.

Research supervision

2026	La Trobe University: primary supervisor of two master's students and co-supervisor of one master's student. University of Melbourne: co-supervisor of one PhD student.
Mar 2024 – May 2026	KAIST: Co-advised and mentored 7 graduate students (PhD/MS) during postdoctoral research. Related student-first-author papers are marked in Publications.

Community and outreach

Jan 2023 – May 2026	Global Shaper, World Economic Forum. Founding Curator of Daejeon Hub; Vice Curator (2023–2024).
Mar 2022 – Dec 2022	Instructor, *Multi-User Game Development with Unity*, KAIST Center for Gifted Education.
Jul 2021 – Aug 2021	Python Coach, NAVER.